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Analyzing Academic and Behavioral Factors for Student Performance Prediction

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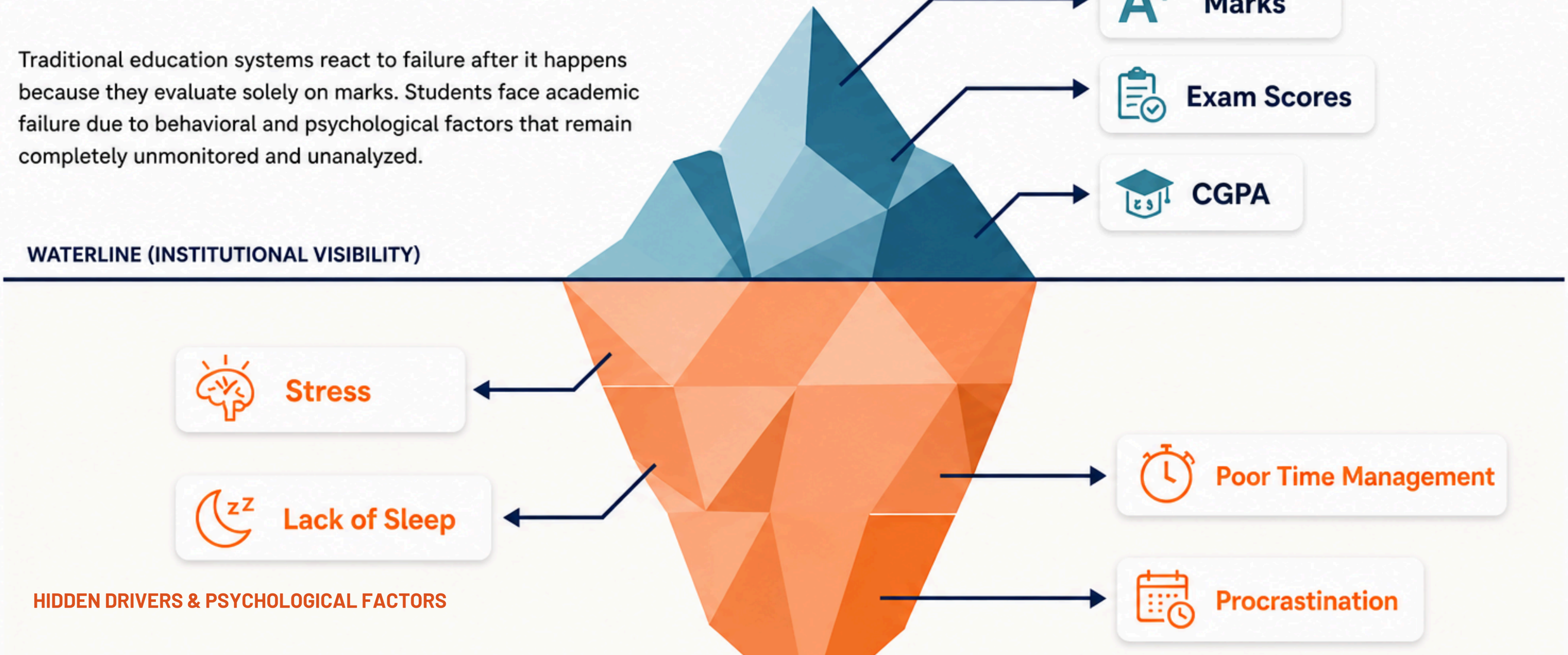
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Traditional Evaluation Misses the Hidden Drivers of Failure

Traditional education systems react to failure after it happens because they evaluate solely on marks. Students face academic failure due to behavioral and psychological factors that remain completely unmonitored and unanalyzed.



Literature Review

[1]

M. Goyal & S. Kumar (2024) - "Early Detection of At-Risk Students Using Machine Learning" - arXiv: 2412.09483

Finding: ML models using academic data can predict at-risk students with high accuracy (up to 87%)

Gap: Does not include stress or procrastination scales; limited analysis of behavioral impact.

[2]

F. Rozental et al. (2022) - "Academic procrastination in university students: Systematic review and meta-analysis" - PMC9480259

Finding: Academic procrastination has a significant negative effect on academic performance across disciplines.

Gap: No predictive model developed; purely statistical/psychological analysis.

[3]

A. S. Dhall et al. (2023) - "Stress and Anxiety Level Prediction in Students Using ML" - IEEE Xplore: 11360699

Finding: ML techniques can predict stress and anxiety levels with accuracy above 85% using survey-based features.

Gap: Focuses only on stress/anxiety; does not link with academic performance or CGPA prediction

[4]

P. Gupta & R. Pathak (2020) - "Multi-factor ML framework for predicting student academic performance" - PMC12642127

Finding: A multi-factor ML framework improves prediction of academic performance compared to single-factor models.

Gap: Does not consider behavioral factors like stress and procrastination in the model

Research Gap

What is missing in existing research?

01 Limited studies combine behavioral psychology (stress + procrastination) with ML for predicting academic performance.

02 Existing models predict outcomes but lack explanation of psychological factors.

03 Few studies focus on early identification of at-risk students using behavioral signals

04 Stress and procrastination have been studied separately in psychology but never jointly engineered as features in a predictive ML model.

This project directly closes all four gaps — combining data science with behavioral psychology.

Motivation / Significance

Dimension	Traditional System	Proposed AI-Based Predictive System
Evaluation Metric	Grades Only	Behavior + Grades
Intervention Timing	Post-Failure / Reactive	Early Detection / Proactive
Diagnostic Capability	No explanation for failure	Reason-based behavioral insights
Scope	Academic records only	Psychology + Academic data

Why This Matters

Early Identify at-risk students before exam failure occurs

Explain Provide behavioral explanations for poor performance

Act Enable targeted academic intervention by faculty

Problem Statement

Core Problem

Students in higher education often face academic failure not due to lack of intelligence, but due to unmonitored behavioral and psychological factors — including stress, procrastination, poor sleep, and irregular attendance. Traditional grading systems only react after failure has occurred, providing no early warning and no explanation of the root cause.

Specific Gaps This Project Addresses:

1. No real-time or early-stage academic risk monitoring system exists for postgraduate students.
2. Behavioral factors (stress, procrastination) are never formally measured or used in academic prediction.
3. No system explains WHY a student is underperforming — it only shows the outcome (low CGPA).

Objectives

01 Analyze Behavioral Factors

Collect and quantify stress and procrastination levels from M.Sc students using validated survey instruments.

02 Develop AI Prediction System

Build a machine learning pipeline using Linear Regression (baseline) to predict student CGPA from behavioral + academic features.

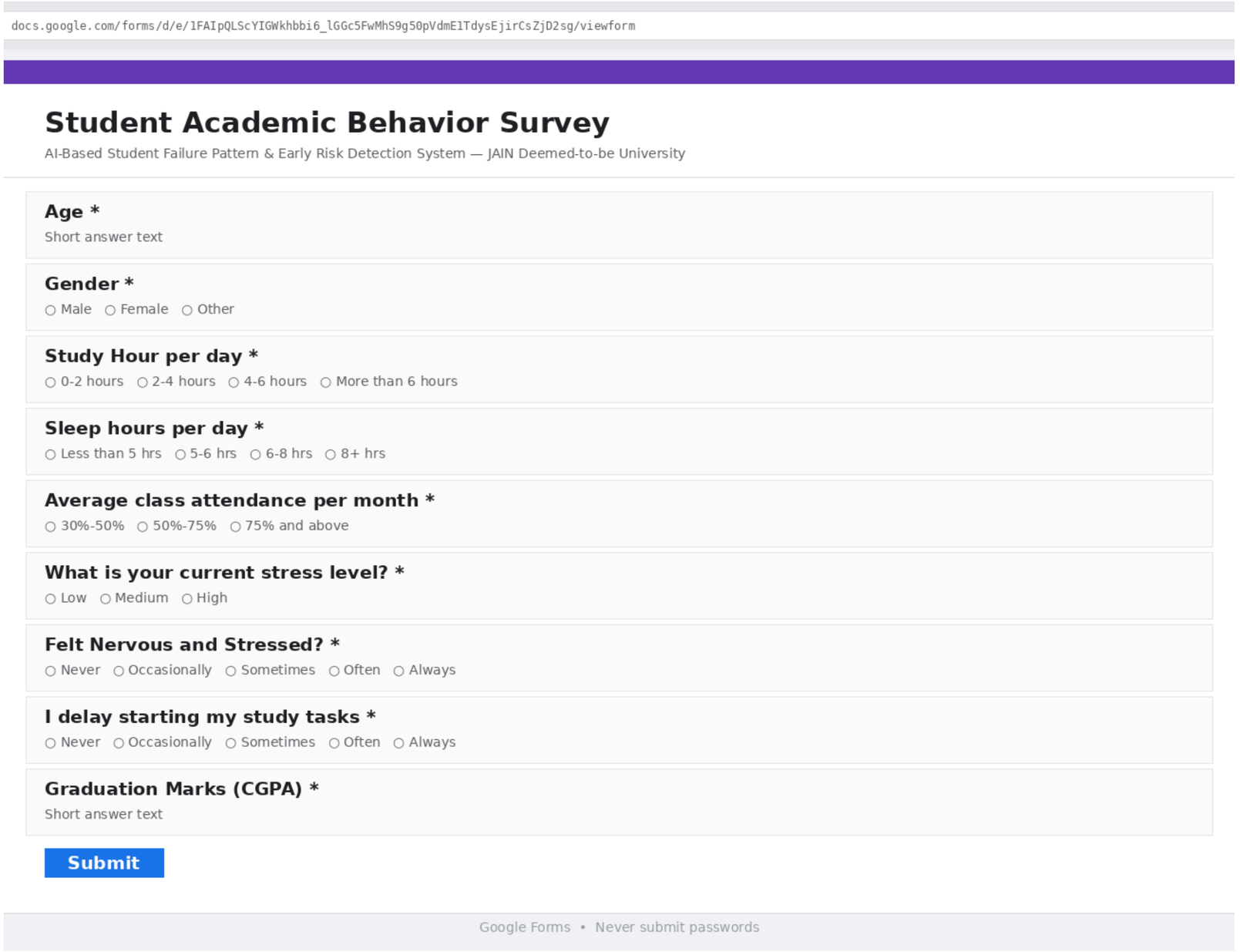
03 Predict Academic Outcomes

Generate predicted CGPA scores and classify students into Low / Medium / High risk categories based on thresholds.

04 Enable Early Intervention

Provide reason-based behavioral insights to faculty so targeted academic support can be given before exam failure occurs.

Data Source & Description — Google Form Survey



52

Responses Collected

29

Survey Fields / Questions

Survey Link:

docs.google.com/forms/d/e/1FAIpQLScYIGWkhbbi6_lGGc5FwMhS9g50pVdmE1TdysEjirCsZjD2sg/viewform

Data Cleaning & Cleaned Dataset — First 10 Records

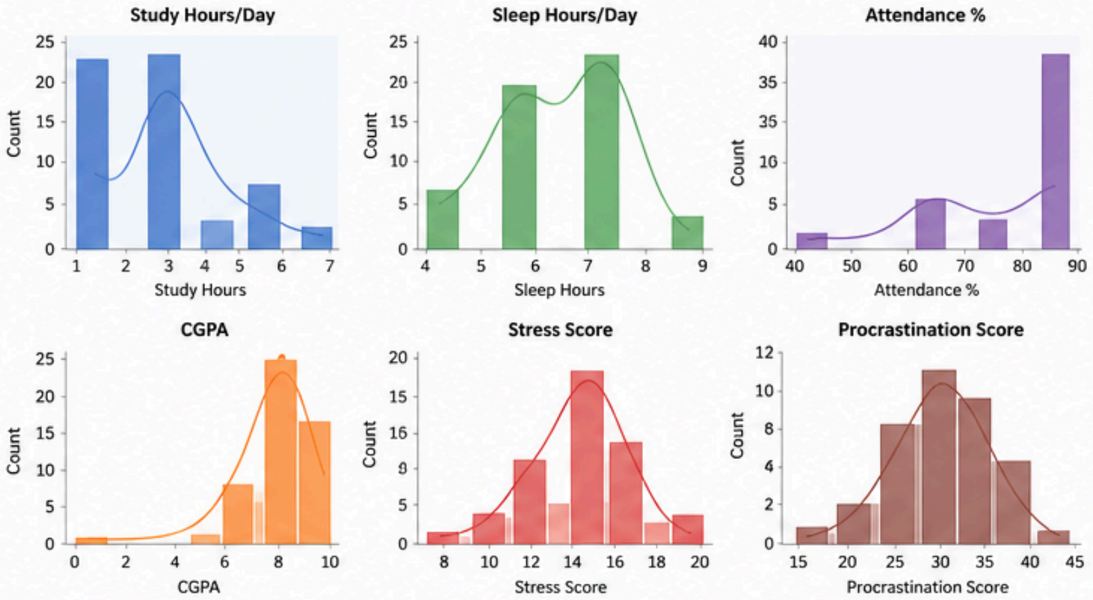
study_hrs	sleep_hrs	attend_%	cgpa	stress_sc	proc_sc	age	gender	semester	marks_10
1	5.5	85	7.4	21	44	24	M	2	9.3
1	5.5	62	8.5	11	33	25	M	2	5.8
3	5.5	85	7.9	12	24	22	F	2	6.5
1	7	85	8	15	33	28	F	2	9
1	9	85	8.9	17	40	22	F	2	9.3
7	9	85	10	17	28	80	F	4	10
1	5.5	85	7.5	12	22	23	F	2	8
3	7	62	8.2	15	36	21	M	2	6.9
1	7	85	8.7	12	39	21	M	4	8.7
3	5.5	85	8.8	16	29	28	F	2	9.5

Standardise	Drop	Impute	Encode	Score
Column names → snake_case	Timestamp, Username, Hobbies	Median for 2 missing values	Ranges → numeric midpoints	Stress (5Q) + Proc (10Q) sums

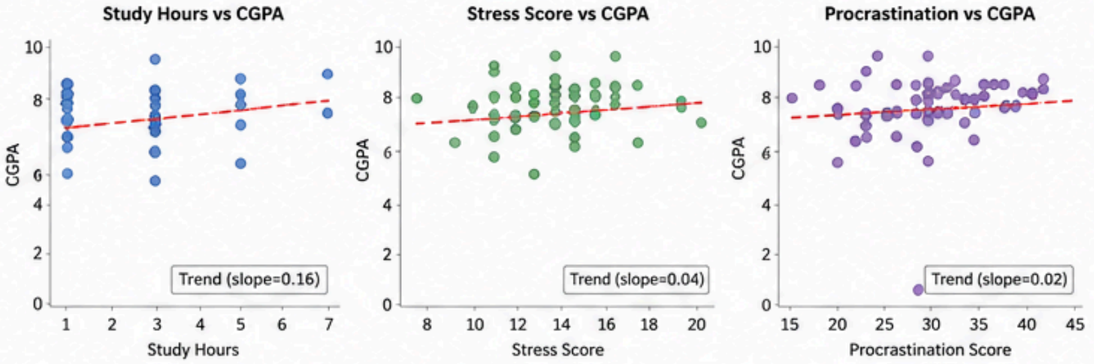
Result: 52 rows × 12 numeric columns | 0 missing values | All features ready for ML

EDA-DISTRIBUTION & FEATURE RELATIONSHIPS

1. FEATURE DISTRIBUTION ANALYSIS



2. FEATURE vs CGPA SCATTER PLOTS

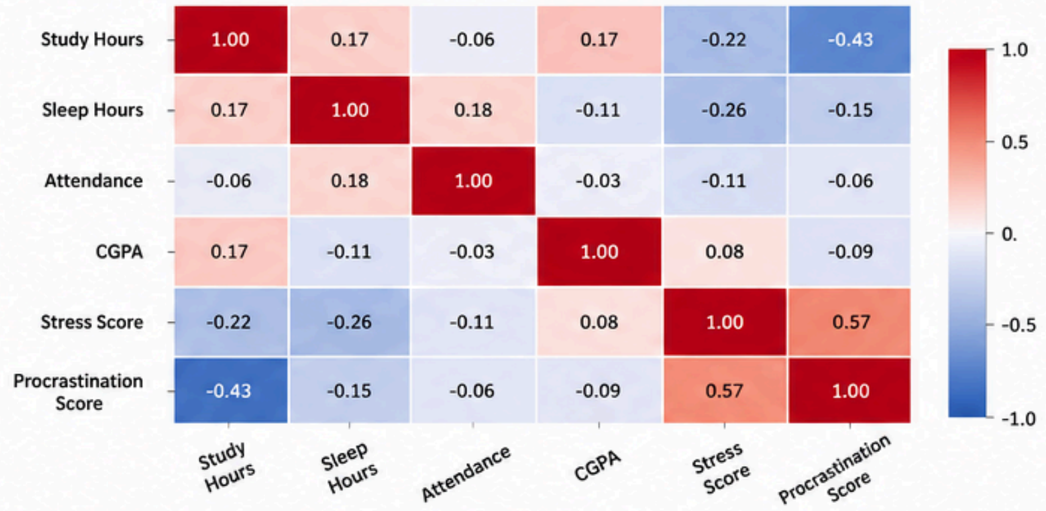


KEY TAKEAWAYS

- Study hours and CGPA show a slight positive relationship.
- Stress score has a weak positive relationship with CGPA.

EDA CORRELATION & STRESS IMPACT

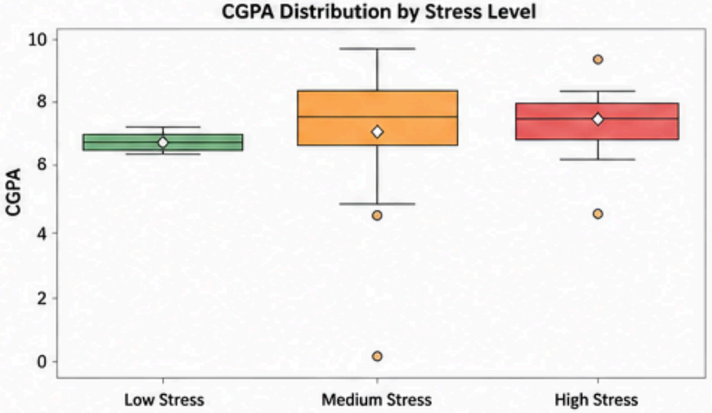
1. FEATURE CORRELATION HEATMAP



KEY TAKEAWAYS

- Study hours have a positive correlation with CGPA.
- Stress score and Procrastination score are strongly positively correlated.
- Procrastination score has a slight negative correlation with CGPA.

2. CGPA BY STRESS LEVEL (BOX PLOT)



KEY TAKEAWAYS

- Students with medium stress have the highest variation in CGPA.
- High stress level is associated with slightly lower CGPA.
- Low stress level shows stable CGPA.

Methodology

1	2	3	4	5	6
Data Collection	**Data Cleaning**	**Feature Engineering**	**EDA**	**Data Preparation**	**Pattern Identification**
Google Forms 52 responses	Encode + impute standardise	Stress Score Proc Score	Scatter plots Heatmap etc	Train-Test Split / Feature Selection	Low/Med/High Flag + explain

Action Plan

Phase 1: Data Expansion

Collect 500+ responses across programmes
Distribute form to multiple batches
Validate expanded dataset quality

Phase 2: Advanced ML Models

Implement Random Forest & Gradient Boosting
Compare MAE, RMSE, R^2 across models
Add classification (Low/Med/High risk label)

Phase 3: Explainability (SHAP)

Add SHAP values to explain predictions
Generate per-student behavioral insight report
Visualize feature contributions

Phase 4: Web App Deployment

Build Streamlit web application
Deploy on Streamlit Cloud (free hosting)
Pilot test with faculty members

Conclusion

"Predictive AI is ultimately a tool for human empathy — by measuring what was previously invisible, we can build education systems that support the whole student."

1. This project proves that behavioral factors — stress and procrastination — are measurable, encodable, and predictive of student CGPA.

2. Behavioral responses were successfully converted into numerical features

3. Stress score and procrastination score were engineering to represent student behavior

4. EDA revealed that higher stress and procrastination negatively impact academic performance
   The dataset is now ready

References (IEEE Format)

[1] M. Goyal and S. Kumar, "Early Detection of At-Risk Students Using Machine Learning," arXiv preprint arXiv:2412.09483, 2024.

[2] F. Rozental, R. Carlbring, and P. Andersson, "Academic procrastination in university students: Systematic review and meta-analysis," PLoS One, vol. 17, no. 3, 2022.

[3] A. S. Dhall, R. Sharma, and S. Singh, "Stress and Anxiety Level Prediction in Students Using Machine Learning," in Proc. IEEE ICOSEC, 2023.

[4] P. Gupta and R. Pathak, "Multi-factor Machine Learning framework for predicting student academic performance," PLoS One, vol. 15, no. 10, 2020.

Thank You

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